Department of Mining Engineering
Indian Institute of Technology (ISM), Dhanbad

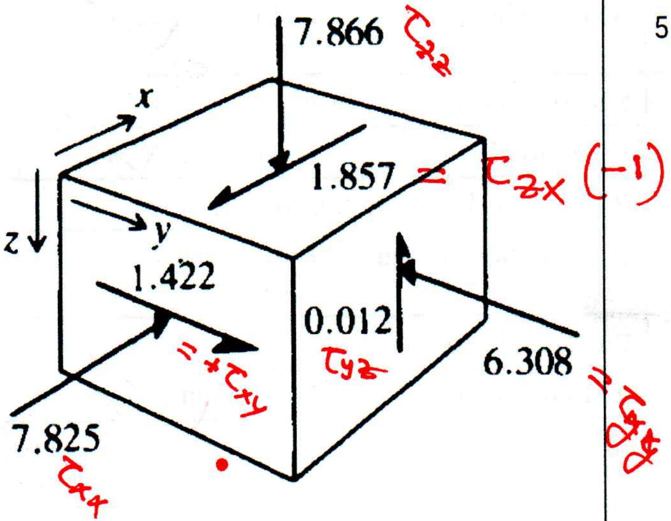
Examination: **MID SEMESTER** | **B. Tech (Open Elective) 6th Semester** | Session: **2025-26**

Subject: **MNO 401 ROCK ENGINEERING** | Semester: **Winter**

Time: 2 **HOUR**

Max. Marks: 30

Instructions:
1. Answer **ALL** questions unless internal choice is specified.
2. Write answers neatly and label all diagrams where relevant. Neat & accurate sketches assists you in solving the problems logically and faster
3. Assume logical data, where necessary
4. Use **Rock Mechanics Sign Convention** to solve the problems
5. Illustrations/diagrams earn credit wherever they add clarity.
6. For multiple choice questions – write the question number, the correct option with its narration in the answerscript

Q. No.	**Question**	**Marks**
1	The normal to a plane in xyz-coordinate system is directed along a line $\dfrac{x-5}{32} = \dfrac{y-2}{65} = \dfrac{z-3}{45}$ and the traction on this plane is $\vec{p} = 32\hat{\imath} + 65\hat{\jmath} + 45\hat{k}$ If the shear component of the tractions on the orthogonal planes aligned with the co-ordinate system are zero, then determine the traction component on the said orthogonal planes aligned with the co-ordinate system.	5
2	The components of the traction on orthogonal planes are shown in the figure on the right. Using the rock-mechanics sign-convention write the matrix of traction components $\begin{bmatrix} \tau_{xx} & \tau_{yx} & \tau_{zx} \\ \tau_{xy} & \tau_{yy} & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & \tau_{zz} \end{bmatrix}$  	5
3	The strength of a coal pillar can be estimated using the formula $$\sigma_p = 7.17\left(\frac{w}{h}\right)^{0.46} h^{-0.20}$$ where $\sigma_p =$ Pillar strength (MPa) $w =$ Pillar width (m) $h =$ Pillar height (m) If the centre-to-centre distance between the square pillars of an underground coal mine at a depth of 400 m is 45 m. Then determine the Factor of Safety of the pillars assuming that (a) the gallery width is 4.5 m (b) unit weight of the overlying strata is $\gamma = 25\ \mathrm{kN/m^3}$ (c) working height of the pillar is 3.0 m	10

Q. No.	Question				Marks
4	Rock masses are best described as				1
	A) CHILE	B) Elastic	C) Plastic	D) DIANE	
5	The Malpasset Dam failed due to				1
	A) Overtopping	B) Structural design flaws	C) Foundation geology issues	D) Earthquake	
6	Spalling occurs due to				1
	A) Low stress	B) Weathering	C) High tangential stress	D) Cooling	
7	Coalbrook disaster involved				1
	A) Gas explosion	B) Pillar failure	C) Flooding	D) Chemical fire	
8	Hysteresis indicates				1
	A) Elastic behavior	B) Energy loss	C) Perfect plasticity	D) Isotropy	
9	Crown pillars provide				1
	A) Ventilation	B) Ore transport	C) Surface stability	D) Drainage	
10	Rockbursts occur due to				1
	A) Low energy release	B) Sudden strain energy release	C) Thermal expansion	D) Water ingress	
11	Spherical projection is used in				1
	A) Hydraulics	B) Wedge analysis	C) Surveying	D) Ventilation studies	
12	Rock mass classifications are part of				1
	A) Analytical design	B) Observational design	C) Empirical design	D) Hydrological design	
13	Block caving is a type of				1
	A) Entry stope	B) Non-entry stope	C) Pillar mining	D) Strip mining	

The relevant formula for solving the problem are given as under.
(The symbols have their usual meaning)

$$(\vec{p}) = [\tau]\,[\hat{n}],$$

$$\tau' = L\tau L^T$$

$$L = \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix} \text{for 2D}$$

$$L = \begin{bmatrix} \cos(\angle x'ox) & \cos(\angle x'oy) & \cos(\angle x'oz) \\ \cos(\angle y'ox) & \cos(\angle y'oy) & \cos(\angle y'oz) \\ \cos(\angle z'ox) & \cos(\angle z'oy) & \cos(\angle z'oz) \end{bmatrix} \text{for 3D}$$

Stress and Traction Problem Solution

Given:
Normal direction ratios = (32, 65, 45)
Traction vector p = 32 i + 65 j + 45 k

Step 1: Magnitude of normal vector
$|n| = \text{sqrt}(32^2 + 65^2 + 45^2)$
$= \text{sqrt}(1024 + 4225 + 2025)$
$= \text{sqrt}(7274)$

Step 2: Unit normal vector components
$n_x = 32 / \text{sqrt}(7274)$
$n_y = 65 / \text{sqrt}(7274)$
$n_z = 45 / \text{sqrt}(7274)$

Step 3: Using Cauchy's formula
$p = \sigma \cdot n$

Since shear stresses on coordinate planes are zero, stress tensor is diagonal:
$\sigma = \text{diag}(\sigma_x, \sigma_y, \sigma_z)$

So,
$\sigma_x (32 / \text{sqrt}(7274)) = 32$
$\sigma_y (65 / \text{sqrt}(7274)) = 65$
$\sigma_z (45 / \text{sqrt}(7274)) = 45$

Step 4: Solving
$\sigma_x = \sigma_y = \sigma_z = \text{sqrt}(7274)$
≈ 85.29

5

Final Answer:
$\sigma_x = \sigma_y = \sigma_z \approx 85.29$

Rock Mechanics Stress Matrix Solution

Sign Convention Used:
- Compression = Positive
- Shear stress is positive when it acts in the negative coordinate direction on a positive face.

Normal Stresses:
tau_xx = ~~6.308~~ 7.825
tau_yy = 1.422 6.308
tau_zz = 7.866 7.866

Shear Stresses (Rock Mechanics Convention):
tau_xy = tau_yx = ~~-0.012~~ +1.422
tau_xz = tau_zx = -1.857 -1.857
tau_yz = tau_zy = -7.825 +0.012

Final Stress (Traction) Matrix:

[6.308 -0.012 -1.857]
[-0.012 1.422 -7.825]
[-1.857 -7.825 7.866]

5

Wrong solution provided by Chat GPT

$$\begin{bmatrix} 7.825 & 1.422 & -1.857 \\ & 6.308 & 0.012 \\ \text{Sym} & & 7.866 \end{bmatrix}$$

Factor of Safety of Coal Pillar (Salamon–Munro Formula)

Given:
Depth of cover, $H = 400$ m
Centre-to-centre spacing, $C = 45$ m
Gallery width, $g = 4.5$ m
Unit weight, $\gamma = 25$ kN/m³
Working height (pillar height), $h = 3.0$ m

Step 1: Pillar Width
$w = C - g = 45 - 4.5 = 40.5$ m

Step 2: Pillar Strength
$\sigma_p = 7.17\,(w/h)^{0.46}\,h^{-0.20}$
$w/h = 40.5 / 3.0 = 13.5$
$\sigma_p \approx 19.1$ MPa ✓ 19.056 ≈ 19.1

Step 3: Vertical Stress
$\sigma_v = \gamma H = 25 \times 400 = 10$ MPa

Step 4: Pillar Load (Tributary Area Method)
$\sigma_{load} = \sigma_v\,(C^2 / w^2)$
$= 10\,(45^2 / 40.5^2)$
$\sigma_{load} \approx 12.34$ MPa ✓

10

Step 5: Factor of Safety
$FOS = \sigma_p / \sigma_{load}$
$FOS \approx 1.55$

Final Answer:
Factor of Safety ≈ 1.55

1.543 ≈

1.55 ✓